Slide 1: AgriQFS: Quantum-Classical Hybrid Feature Selection for Temporal UAV Imagery

- Presented By
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Slide 2: Motivation

• We use UAV orthomosaic RGB imagery to monitor cotton before and after defoliation.

• The goal is to distinguish pre-defoliation canopy from post-defoliation exposed structures.

• This matters because defoliation timing affects harvest readiness, input decisions, and crop quality.

• The challenge is that field imagery changes with texture, canopy density, soil visibility, illumination, and scene complexity.

• So, the core problem is not only classification but finding robust features under real agricultural variability.

• Accurate cotton defoliation monitoring is difficult because UAV imagery captures real field variability, not clean laboratory conditions.

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Slide 3: Research Contribution

• This study is grounded in interpretable feature descriptors extracted from cotton UAV imagery, capturing spectral, textural with defoliation.

• It introduces a

• variational quantum circuit (VQC)

• to search for compact feature subsets that preserve discriminative information.

• The method evaluates whether

• interaction-aware quantum-inspired selection

• can outperform standard classical feature-selection baselines.

• The selected subset is then used by a

• classical Support Vector Machine with a Radial Basis Function (RBF) kernel

• for final prediction.

• The main contribution is a

• hybrid quantum-classical pipeline

• that improves interpretability, keeps deployment practical, and studies quantum ideas in an agricultural remote-sensing setting.

• This work proposes a hybrid feature-selection framework that combines interpretable agricultural descriptors with quantum-emulated search.

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Slide 4: Why Quantum Computing?

• Before understanding more about quantum computers, we need to understand the concepts of quantum physics.

• Quantum physics puts everything into question. It defies every intuition which we have about our natural world.

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• Figure 1: view of quantum-enhanced search

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Slide 5: Superposition

• Classically, a state is this or that.

• In quantum mechanics, a state can be this and that until it is measured.

• That makes superposition a powerful idea for exploring many candidate solutions at the same time.

• A quantum system can hold multiple possible states in play until measurement forces a single outcome

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• Figure 2:

• Single-qubit state on the Bloch sphere

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Slide 6: Entanglement

• Classically, we often rank one feature at a time.

• .

• Quantum systems can encode relationships between parts of the system directly.

• That is why entanglement is the most natural bridge to interaction-aware feature selection.

• Two quantum states can become linked so strongly that they stop behaving like independent objects.

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• Figure 3: Correlated two-qubit Bell state

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Slide 7: Tunneling

• Classical optimization can get stuck behind a barrier in the search.

• landscape.

• Quantum behavior allows probability to appear on the other side.

• of that barrier.

• That is why tunneling is often used as the mental model for better global search.

• Quantum systems can cross barriers that would trap a purely classical search process

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• Figure 4: Quantum tunneling through an energy barrier

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Slide 8: Overview of the Hybrid Quantum-Classical Feature-Selection Pipeline

• A quantum-emulated feature-selection stage is coupled with a classical classifier to balance interaction-aware search with practical inference

• Figure 5: Overview of the Hybrid Quantum-Classical Feature-Selection Pipeline

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Slide 9: Hybrid Quantum-Classical Approach

• This research work uses a

• variational quantum circuit (VQC)

• as a

• feature selector.

• The quantum part is simulated on a classical computer using

• Python and Qiskit on macOS.

• This lets us test

• superposition, entanglement, and interaction-aware search

• in a reproducible way without requiring real quantum hardware.

• The selected subset is then passed to a

• classical Support Vector Machine with a Radial Basis Function (RBF) kernel

• , which keeps the final pipeline practical and deployable.

• So the contribution is a

• hybrid workflow

• : quantum-emulated search for better feature subsets, classical learning for stable prediction.

• We do not run on a physical quantum computer, we emulate the quantum circuit in Python to study whether quantum-inspired feature interactions improve feature selection

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Slide 10: Why the Hybrid Architecture Is Effective

• The framework assigns each paradigm to the stage where it contributes most: quantum modeling for subset discovery, classical learning for stable inference

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Slide 11: Why the Method Is Most Reliable

• AgriQFS is most adaptative when the task demands compact, interpretable feature selection under challenging imaging conditions

• Interaction-Dominated Regimes:

• Its advantage increases when useful information is distributed across interacting descriptors rather than isolated within a single feature.

• Perturbed Imaging Conditions:

• under

• glare

• and

• additive sensor noise

• , where stable subset selection becomes more challenging. This aligns with robustness under field variability.

• Interpretability for Domain Science

• The final subset remains understandable in agronomic and remote-sensing terms.

• That makes the method easier to analyze scientifically than an opaque end-to-end representation.

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Slide 12: The Role of the 4-Qubit Configuration

- the 4-qubit configuration provides the clearest balance between compact feature selection, preserved class structure, and practical interpretability
- What 4 qubits mean here
- The 4-qubit setting corresponds to selecting a compact subset of descriptors for downstream classification.
- T
- his regime produces the most interpretable and practically useful subset.
- Why this regime is important
- Too small a subset can lose important structure.
- Too large a subset reduces compactness and weakens the argument for efficient selection.
- The 4-qubit case offers the best ground
- for experimentation
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- Figure 6:
- Qubit State Duality

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Slide 13: Why a Compact Subset Still Works

• PCA of the selected descriptors shows that discriminative structure is preserved without relying on the full feature set

• Figure 7: PCA projection of the QFS-selected k=4 feature space

• Class Structure Preserved

• The selected subset retains visible separation between pre- and post-defoliation states in the projected feature space.

• Compact but Informative

• This separation is achieved with a small descriptor set, indicating that the selected features preserve discriminative information efficiently.

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Slide 14: Interpretable 14-D Feature Representation

• The framework operates on physically meaningful descriptors derived from UAV imagery, capturing spectral response, texture organization, and exposed-structure geometry relevant to cotton defoliation

• Vegetation Indices

• These descriptors capture canopy color shifts associated with senescence, exposure, and boll visibility.

• They provide the spectral foundation for separating pre- and post-defoliation field states.

• ExG, visible-band NDVI, and RBR.

• GLCM Texture

• Texture descriptors quantify spatial organization in the canopy, including smoothness, contrast, and structural regularity.

• These are important because defoliation changes not only color, but also the spatial pattern of exposed and vegetated regions.

• Correlation, Contrast, Energy, Entropy, and Homogeneity.

• Morphological / Structural Descriptors

• Structural descriptors summarize the geometry of exposed regions after canopy change becomes visible.

• They complement spectral and textural features by encoding shape-level variation tied to exposed cotton structures.

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Slide 15: Quantum Encoding

• Candidate feature subsets are mapped into a 4-qubit representation so that descriptor interactions can be expressed within the quantum state before scoring

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• Input

• Candidate subset of descriptors from the 14-D feature space

• Feature Map

• Classical descriptor values are embedded using the

• ZZFeatureMap

• Interaction Encoding

• Pairwise relationships are introduced into the quantum phase

• .

• Prepared State

• The encoded state becomes the input to the variational circuit.

• Notation meaning

• .

• : the classical input vector, here the selected feature descriptor values

• :

• the quantum feature map that encodes the classical input into a

• qua.ntum

• state.

• :

• the initial 4-qubit state, meaning all four qubits start in the

• state.

• : tensor product repeated over 4 qubits.

• :

• the classical feature vector is transformed into an encoded 4-qubit quantum state.

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Slide 16: Pre- and Post-Defoliation Field Scenes with Detected Cotton Regions

• UAV imagery shows the field before and after defoliation, alongside the corresponding detected cotton regions used for scene-level qualitative assessment

• Figure 8: (a)

• Pre-defoliation UAV scene,

• (b)

• detected cotton regions before defoliation,

• (c)

• post-defoliation UAV scene,

• &

• (d)

• detected cotton regions after defoliation

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Slide 17: Robustness Across Imaging Conditions

- The table compares subset-selection methods under fog, glare, shadow, and additive noise to evaluate how stable each selected feature
- Method
- Fog
- Glare
- Shadow
- Noise (σ = 0.15)
- AgriQFS (QFS-k4)
- 0.8301
- 0.6280
- 0.7699
- 0.4903
- MRMR (MI-k4)
- 0.9054
- 0.5935
- 0.9634
- 0.4624
- RFE-k4
- 0.8667
- 0.6237
- 0.7699
- 0.4516
- NonExG-k4
- 0.7333
- 0.5742
- 0.7699
- 0.4602
- AgriQFS (QFS-k4):
- Agricultural Quantum Feature Selection, using a quantum-emulated search to select 4 features.
- MRMR (MI-k4):
- Minimum Redundancy Maximum Relevance, a mutual-information filter baseline with 4 features.
- RFE-k4:
- Recursive Feature Elimination, a wrapper-based 4-feature baseline.
- NonExG-k4:
- an ablation baseline excluding the Excess Green cue to test subset dependence.

• Our framework is more effective because it performs interaction-aware subset selection, allowing complementary spectral and textural cues to be chosen jointly

• Table 1 :Comparative Robustness of Feature-Selection Methods

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Slide 18: Robustness Landscape Across Perturbed Imaging Conditions

• The heatmap compares subset-selection strategies across clean and perturbed conditions to reveal where quantum-guided feature selection remains most competitive under field-relevant stress

• Figure 9: Condition-Wise Accuracy of QFS,

• NonExG

• , and Mutual-Information Baselines

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Slide 19: Conclusion

• The study shows that quantum-guided subset selection can produce compact, interpretable feature sets for cotton defoliation assessment from UAV imagery

• This work presented a

• hybrid quantum-classical framework

• for feature selection in cotton defoliation monitoring from UAV imagery.

• The method operates on

• interpretable spectral, textural, and structural descriptors.

• The results show that

• AgriQFS identifies a compact subset

• that preserves discriminative structure between pre- and post-defoliation states.

• Under clean conditions, the framework achieves strong performance while also maintaining

• low false-positive behavior

• , which is important for field reliability.

• Under perturbed conditions, the method is

• not uniformly best across every scenario

• , but it remains competitive and shows its clearest value in regimes where

• interaction-aware selection

• matters.

• Overall, supports the idea that

• quantum-emulated feature selection is scientifically useful as a subset-discovery mechanism

• , even when final inference remains classical method.

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Slide 20: Future Work

- Extend the framework from the current descriptor space to
- larger and more diverse feature sets
- , including multispectral or hyperspectral imaging data.
- Evaluate the method across
- additional seasons, fields, cultivars, and acquisition conditions
- to test generalization beyond the current study setup.
- Investigate whether the framework can support
- finer-grained defoliation estimation
- , mo
- re than
- pre-/post-defoliation discrimination.
- Integrate the method into a broader
- precision-agriculture LLM decision pipeline
- , where robust feature selection supports prescription mapping and field-scale monitoring
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Slide 21: Publications and Accepted Work

• H. Manjunatha, P. Sundaravadivel, S. Borah, L. Tamil, P. Knight, H. Torbert, S. Kumpatla, "Beyond Green: Domain-Aware Self-Validated Instance Counting for Loropetalum and Non-Green Ornamental Species using Annotation-Free Robustness Metrics", In Proceedings of CVPR, Jun 2026. Accepted

• A. Kode, H. Manjunatha, S. S. Borah, H. A. Torbert, P. Sundaravadivel, "Scalable Voice-Controlled Drone Autonomy: From Edge-Deployed Single Agents to Distributed HITL Swarm Simulations," Accepted DCAS 2026.

• H. Manjunatha, S. Borah, A. Anand, P. Sundaravadivel, L. Tamil, S. Kumpatla, P. Knight, H. Torbert, "Comparative Analysis of Large Language Models for Multimodal Agricultural Intelligence in Precision Nursery Management Systems", In Proceedings of LLM Conference, Dec 2025. Accepted LLM conference 2025.

• P. Sundaravadivel, H. Manjunatha, K. Narasimhamurthy, et al., "Ros-AI: An LLM-Enhanced Scalable Multimodal Framework for UAV-Based Rose Bloom Analysis in Precision Agriculture", TechRxiv Preprint, Oct 2025. DOI: 10.36227/techrxiv.176108108.86171044/v1.

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Slide 23: Thanks for listening!

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